

Database Name: LibraryDB

Step 1: Create the following tables with appropriate constraints

1. Books

- book_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- title (VARCHAR(100), NOT NULL)
- author_id (INT, FK, NOT NULL)
- genre (VARCHAR(50))
- published_year (YEAR)

2. Authors

- author_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- name (VARCHAR(100), NOT NULL)
- country (VARCHAR(50))

3. Borrowers

- borrower_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- name (VARCHAR(100), NOT NULL)
- join_date (DATE)

4. BorrowingRecords

- record_id (INT, PK, AUTO_INCREMENT, NOT NULL)
- book_id (INT, FK, NOT NULL)
- borrower_id (INT, FK, NOT NULL)
- borrow_date (DATE)
- return_date (DATE)

Step 2: Insert the following sample data

author_id	name	country
1	Andy Weir	United States
2	Emily St. John Mandel	Canada
3	Hilary Mantel	United Kingdom
4	Peter Frankopan	United Kingdom
5	Liu Cixin	China

book_id	title	author_id	genre	published_year
1	Project Hail Mary	1	Science Fiction	2021
2	The Martian	1	Science Fiction	2011
3	Artemis	1	Science Fiction	2017
4	Sea of Tranquility	2	Science Fiction	2022
5	Station Eleven	2	Science Fiction	2014
6	Wolf Hall	3	History	2009
7	Bring Up the Bodies	3	History	2012
8	The Silk Roads	4	History	2015
9	The Future of War	4	History	2024
10	The Three-Body Problem	5	Science Fiction	2008

borrower_id	name	join_date
1	Alice Johnson	2023-03-15
2	Rahim Uddin	2022-07-20
3	Sarah Kim	2024-01-05
4	Jamil Hasan	2021-11-30
5	Nadia Chowdhury	2025-02-28

record_id	book_id	borrower_id	borrow_date
1	1	1	2024-08-05
2	2	1	2024-09-12
3	3	1	2024-12-01
4	4	2	2023-01-10
5	5	2	2023-05-12
6	10	2	2024-06-11
7	6	3	2025-04-01
8	7	3	2025-05-10
9	8	4	2022-11-10
10	9	5	2025-06-10

Step 3: Write queries to solve the following

- Q1. Show the names of authors who have written more than 2 books in the genre 'Science Fiction'.
- Q2. List all books that were never borrowed.
- Q3. Show each borrower's name and how many books they have borrowed so far.
- Q4. Find the top 2 most borrowed books.
- Q5. Show the list of borrowers who borrowed books authored by authors from 'United Kingdom'.
- Q6. List all books in the 'History' genre published after the year 2010.
- Q7. Find all books that have been borrowed more than 5 times.
- Q8. List borrowers who borrowed books before 2025 but haven't borrowed anything in 2025 or later.

Solution:

Ans.1

```
SELECT a.name
FROM Authors a
JOIN Books b ON a.author_id = b.author_id
WHERE b.genre = 'Science Fiction'
GROUP BY a.author_id, a.name
HAVING COUNT(b.book_id) > 2;
```

name

Andy Weir

Ans.2

```
SELECT b.title
FROM Books b
LEFT JOIN BorrowingRecords br ON b.book_id = br.book_id
WHERE br.book_id IS NULL;
```

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0005 seconds.)

```
SELECT b.title FROM Books b LEFT JOIN BorrowingRecords br ON b.book_id = br.book_id WHERE br.book_id IS NULL;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

title

Ans.3

```
SELECT bo.name, COUNT(br.book_id) AS books_borrowed
FROM Borrowers bo
LEFT JOIN BorrowingRecords br ON bo.borrower_id = br.borrower_id
GROUP BY bo.borrower_id, bo.name;
```

name	books_borrowed
Alice Johnson	3
Rahim Uddin	3
Sarah Kim	2
Jamil Hasan	1
Nadia Chowdhury	1

Ans.4

```
SELECT b.title, COUNT(br.record_id) AS borrow_count
FROM Books b
JOIN BorrowingRecords br ON b.book_id = br.book_id
GROUP BY b.book_id, b.title
ORDER BY borrow_count DESC
LIMIT 2;
```

title	borrow_count ▼ 1
Bring Up the Bodies	1
Sea of Tranquility	1

Ans.5

```
SELECT DISTINCT bo.name
FROM Borrowers bo
JOIN BorrowingRecords br ON bo.borrower_id = br.borrower_id
JOIN Books b ON br.book_id = b.book_id
JOIN Authors a ON b.author_id = a.author_id
WHERE a.country = 'United Kingdom';
```

name
Sarah Kim
Jamil Hasan
Nadia Chowdhury

Ans.6

```
SELECT title
FROM Books
WHERE genre = 'History' AND published_year > 2010;
```

	title
lete	Bring Up the Bodies
lete	The Silk Roads
lete	The Future of War

Ans.7

```
SELECT b.title, COUNT(br.record_id) AS borrow_count
FROM Books b
JOIN BorrowingRecords br ON b.book_id = br.book_id
GROUP BY b.book_id, b.title
HAVING borrow_count > 5;
```

Ans.8

```
SELECT bo.name
FROM Borrowers bo
WHERE bo.borrower_id IN (
    SELECT br.borrower_id
    FROM BorrowingRecords br
    WHERE YEAR(br.borrow_date) < 2025
)
AND bo.borrower_id NOT IN (
    SELECT br.borrower_id
    FROM BorrowingRecords br
    WHERE YEAR(br.borrow_date) >= 2025
);
```

	name
 Delete	Alice Johnson
 Delete	Rahim Uddin
 Delete	Jamil Hasan